

PROFESSIONAL SUMMARY

- As a Senior Java Full Stack Developer about 10 years of professional experience in **Java/J2EE** web application design, leading development, maintenance, and support across all phases of the software development lifecycle.
- Configured Recommendation Service with **Spring Boot**, employing batch and real-time processing to dynamically serve personalized product suggestions to over 500,000 users daily.
- Integrated Spring framework (MVC/IOC/ORM/JDBC/AOP/BATCH/WEBFLOW/CONTEXT) and Struts.
- Proficient in project management and documentation using **JIRA, Confluence, Swagger, and Grafana**
- Utilized Spring Cloud **Netflix Eureka** for service discovery, ensuring microservices dynamically register and locate each other, maintaining 99.99% system availability.
- Optimized build processes using **Maven** by configuring project Object Model (POM) files for efficient dependency management and modular builds, reducing build times by over 30%.
- Architected the backend using **Java** and **Spring Boot** with **Python** scripts for automation, focusing on microservices architecture for scalability and utilizing **Python's** powerful data handling capabilities.
- Configured and implemented **Camunda** workflows to enhance process automation, integrating **Camunda** with microservices to streamline operations and optimize process orchestration efficiency across diverse banking and healthcare applications.
- Designed dynamic decision models using **Camunda BPM** and **DMN**, enabling real-time decision-making aligned with business rules. Configured **Camunda** REST APIs to ensure seamless integration and operational transparency.
- Enhanced system reliability by implementing comprehensive error handling in **Node.js**, utilizing **Express.js** middleware for managing errors and corrective actions.
- Employed **Spring Cloud** Gateway and **Spring Security** for efficient **API** routing, managing secure token validations, and providing a resilient communication channel, reducing API response times by 30%.
- Implemented **Angular 5** services to connect the web application to back-end APIs for data POST/GET operations, enhancing API integration and data exchange reliability.
- Designed and developed a dynamic frontend using **React.js** and **Redux**, creating a responsive, component-based architecture that supports complex state management and real-time data updates.
- Integrated **Hibernate** with **MySQL** for ORM capabilities, efficiently managing transactional data across over 10,000 daily transactions.
- Streamlined **JDBC** and **Hibernate** configuration in Spring Boot for robust database management, integrating **MySQL** and **Oracle** for enhanced data retrieval efficiency.
- Implemented **SQL Server** high availability groups for continuous data availability in financial services, reducing downtime risks.
- Configured **Redis** to enhance caching strategies, which improved response times by 70%
- Configured **Kafka** producers and consumers to ensure reliable message delivery and fault tolerance,
- Configured **IBM MQ, RabbitMQ** configurations for reliable asynchronous message queuing
- Enhanced application security and authentication using **OAuth 2.0** through **AWS API Gateway**, protecting data transfers via **HTTPS**.
- Configured security headers and **CORS** policies in **AWS API Gateway** to enhance system security and prevent web vulnerabilities.
- Implemented disaster recovery strategies across **GCP** and **Azure**, using cross-cloud data replication
- Utilized **Docker** within CI/CD pipelines for consistent environment management across development, testing, and production phases.
- Orchestrated **Jenkins** for CI/CD pipelines, automated integration, and deployment processes, and employed **Docker** and **Kubernetes** for containerization and orchestration.
- Implemented comprehensive testing strategies across the stack using **JUnit, Postman, Mockito, React Testing Library, Jest, Cucumber** and **Cypress**
- Configured **Logstash** to parse and enrich logs before loading into **Elasticsearch**, enhancing system monitoring capabilities by 40%.

- Configured **Apache CXF** within Spring Boot to expose **RESTful** and **SOAP** web services, broadening service interoperability.
- Strong Core Java background with experience in **Collections, Multi-Threading, Java 17, Interfaces, RMI, Serialization, Synchronization, Exception Handling, Java Swing, OOPs techniques.**

TECHNICAL SKILLS

Programming Languages: C, C++, Java 17, SQL, PL/SQL, Python, JavaScript, TypeScript

Java/J2EE Technologies: Servlets, Spring, JSP, JSTL, JDBC, RMI, EJB, JFC/Swing, AWT, Applets, Multi-threading, Java Networking

Frameworks: Spring, Hibernate, Angular JS, Spring Security, Spring Boot, Spring MVC, Spring Data JPA, Spring Cloud (Netflix, Gateway), NgRx, Redux-Saga, Grunt

Web Technologies: JSP, JavaScript, jQuery, AJAX, Angular JS, Angular 2+, XML, HTML5, CSS3, ReactJS, Backbone.JS, Node JS, Express JS, Redux, Tailwind CSS, Webpack, Babel

Relational Database: Oracle 11g/12C, SQL server 2012/2014/2016, MySQL, PostgreSQL

NoSQL Databases: MongoDB, Cassandra, DynamoDB, Redis

Web Services: SOAP 1.2, REST 2.0, JAX-WS, JAX-RPC, JAX-RS, Swagger, RESTful, GraphQL

Application/Web Servers: Apache Tomcat 8.0, Oracle/BEA WebLogic, Apache CXF, IBM WebSphere

Messaging: RabbitMQ, IBM MQ, Apache Kafka

IDEs: Eclipse, IntelliJ 2017.1, Spring Tool Suite (STS) 3.9.3, Atom, PyCharm, Visual Studio

Cloud Environments: Azure, AWS (Lambda), GCP

Methodologies: Agile (Safe 6.x), waterfall, TDD (Test-Driven-Development), Scrum

Other Deployment tools: Netflix Eureka, Kubernetes, Elastic Beanstalk, Docker, OpenShift, MLops (MLairflow, DVC), RedHat, Openshift

Testing technologies/tools: JUnit, Postman, TestNG, Mockito, Cucumber, JMeter, RFT, React Testing Library, Cypress, Jasmine, Karma, LEX

Version Control: Git, Bitbucket, SVN, Rational ClearCase,

Build Tools: Maven 3.3.9, Gradle 3.4.1, Ant 1.10, Jenkins

Platforms: Windows 10, UNIX, Linux

Security Technologies: OAuth2, JWT, Spring Security, HTTPS, TLS, AES

Performance and Monitoring Tools: Prometheus, Grafana, AWS CloudWatch, JVisualVM, Log4J, Logstash, Elasticsearch, Kibana, JIRA

PROFESSIONAL EXPERIENCE

Role: Sr. Full Stack Developer

July 2021 to Present

Client: Bank of America, Charlotte, NC

Responsibilities:

- Leveraged **Spring MVC** in conjunction with **Springfox** to integrate **Swaggers**, increasing API documentation usability by 25%, facilitating interactive testing and automated documentation generation
- Architected the backend using **Java** and **Spring Boot**, focusing on microservices architecture for scalability and utilizing immutability features to enhance application reliability.
- Launched and structured an **Angular 16** project using the **Angular CLI**, creating a modular architecture that facilitated easier maintenance and scalability of the front-end framework, reducing setup time.
- Implemented robust two-way data binding in **Angular 16**, ensuring that UI and data model are synchronized in real-time, reflecting backend changes immediately.
- Optimized **Angular 16** application performance by implementing lazy loading and dependency injection, reducing initial load time by 30% for a large-scale application.
- Configured Spring Security with **OAuth2** authentication, **JWT**, and **HTTPS** and **TLS** for secure data transmission, reducing security breaches by over 30%.
- Designed and implemented workflows using **Camunda**, enabling seamless integration with microservices and improving business process automation efficiency by 30% across multiple banking applications.

- Developed RESTful APIs to interact with **Camunda** for dynamic workflow modifications, facilitating flexible process updates and improving end-user satisfaction in banking operations.
- Leveraged **Spring Data JPA** for robust data interactions using the repository pattern and **JPQL** for custom queries, improving data retrieval speed and storage efficiency by 25%.
- Led the upgrade of Adobe Experience Manager (AEM) from version 6.1 to 6.3, enhancing system stability and introducing new features that improved content management workflows.
- Utilized **Bootstrap** to implement responsive design principles, ensuring the application's compatibility across various devices and screen sizes.
- Leveraged **Spring MVC** in conjunction with **Springfox** to integrate **Swaggers**, increasing API documentation usability by 25%, facilitating interactive testing and automated documentation generation
- Architected the backend using **Java** and **Spring Boot**, focusing on microservices architecture for scalability and utilizing immutability features to enhance application reliability.
- Configured Spring Security with **OAuth2** authentication, **JWT**, and **HTTPS** and **TLS** for secure data transmission, reducing security breaches by over 30%.
- Leveraged **Spring Data JPA** for robust data interactions using the repository pattern and **JPQL** for custom queries, improving data retrieval speed and storage efficiency by 25%.
- Developed **RESTful** APIs using **Spring MVC** for efficient controller and request handling, facilitating seamless API integration and data exchange.
- Implemented **Aspect-Oriented Programming (AOP)** with Spring Cloud, including Spring Cloud Gateway, to achieve separation of concerns and enhance modularization.
- Managed transactional data with **Oracle** and configured **Couchbase** to store and retrieve exchange rates.
- Employed **Redis** for caching frequently accessed data, reducing load on primary databases and enhancing application response times.
- Integrated **CouchBase** with Apache **Kafka** for real-time data streaming, enhancing data sync across systems
- Spearheaded the adoption of **Couchbase** Exadata to optimize large-scale data warehousing projects, enhancing query performance by over 60% for complex data analysis tasks.
- Maintained secure data transmission with **HTTPS** and **TLS**, implemented **AES** encryption for data-at-rest security, adhering to **PCI-DSS** and **GDPR** compliance standards.
- Set up **Prometheus** for metrics collection and monitoring and used **Grafana** for dashboarding and alert management, improving system performance monitoring by over 50%.
- Conducted unit testing with **JUnit** and **Mockito**, achieving a 95% code coverage that ensured the robustness and functionality of the platform.
- Implemented **SLF4J** and **Logback** for comprehensive logging, enhancing system diagnostics
- Integrated **RabbitMQ** for asynchronous messaging, and supported multicurrency functionalities with Forex integration, ensuring the platform meets diverse global needs.
- Led the adoption of **GraphQL** in banking applications, enabling efficient and flexible data querying and manipulation.
- Utilized Apollo Server and **GraphQL** to build and deploy GraphQL APIs, ensuring high performance and scalability in banking applications.
- Managed **Kubernetes** deployments using ConfigMaps and Secrets, ensuring dynamic configuration and efficient resource scaling with Horizontal Pod Autoscalers.
- Created and managed **CI/CD** pipelines in **Jenkins**, automating the build, test, and deployment processes, resulting in a 40% reduction in release times.
- Integrated **Jenkins** with **Git** version control, ensuring that code changes triggered automated builds and unit tests for every commit.
- Utilized **Terraform** to define infrastructure as code for **GCP** services, ensuring consistent environment setups across development, staging, and production.
- Implemented automated CI/CD pipelines using **Terraform** and **Cloud Build**, accelerating release cycles and improving code quality through continuous integration and deployment.
- Architected and deployed complex infrastructure on **GCP** using **Terraform**, enabling automated provisioning and management of cloud resources, which reduced deployment times by 40%.
- Developed and optimized Java-based microservices deployed on **Google Kubernetes Engine (GKE)**, ensuring high availability and scalability for mission-critical applications.
- Implemented serverless solutions using **Google Cloud Functions**, streamlining event-driven workflows and reducing operational overhead by 30%.
- Managed containerized applications using **Google Kubernetes Engine (GKE)**, ensuring high availability and efficient resource utilization for KeyBank's online banking platforms.
- Implemented CI/CD pipelines with GitHub Actions, automating build, test, and deployment processes.

- Created and curated project documentation on GitHub, enhancing project visibility and accessibility.
- Configured GitHub security features to identify and remediate security vulnerabilities in code repositories.
- Applied **Hibernate** for object-relational mapping and input sanitization, enhancing security and data integrity within the platform

Environment: Java 17, Angular 16, Cypress, AEM, Python, Redis, TypeScript, NgRx, Bootstrap, Angular Material, HTML5, CSS3, Spring Boot, Camunda BPM, Microservices, Spring Security, OAuth2, JWT, HTTPS, TLS, Spring Data JPA, JPQL, Hibernate ORM, Spring MVC, Spring, RESTful APIs, Oracle, MySQL, Jenkins, Docker, Kubernetes, GCP Services (BigQuery, DataFlow, GKE, Cloud Functions), Terraform, Git, Github, SonarQube, Netflix Eureka, Prometheus, Grafana, Spring Boot Actuator, JUnit, Mockito, SLF4J, Logback, RabbitMQ, AWS CDK, GraphQL.

Role: Sr. Full-Stack Developer

January 2020 to June 2021

Client: Cigna Healthcare, Bloomfield, CT

Responsibilities:

- Developed and maintained robust server-side applications using Java, employing best practices and advanced features of **Java 8** such as lambdas, streams, and functional interfaces
- Configured **Camunda BPM** to manage complex workflows, ensuring seamless integration with microservices and enhancing the overall process orchestration efficiency across various healthcare systems.
- Enhanced monitoring and debugging of **Camunda** processes using its cockpit tool, allowing real-time tracking and ensuring quicker issue resolution.
- Implemented dynamic imports in **React** applications with **React.lazy** and **Suspense**, optimizing code splitting and improving load times for better user experience.
- Optimized client-side navigation by integrating **React Router**, enabling dynamic routing across the single-page application without full-page reloads.
- Enhanced the application's feature set by integrating a robust reporting dashboard using **React Chart.js**, providing real-time analytics and insights to business users.
- Implemented **React** features like lazy loading for images and non-essential resources to minimize initial page.
- Streamlined the process of feature engineering and data preprocessing by implementing automated pipelines using Apache Airflow.
- Employed **Springfox** to integrate Swagger with **Spring MVC**, enhancing API documentation usability by 25% with automated generation and interactive testing capabilities.
- Scaffolded new microservices efficiently using **Spring Initializr**, including essential dependencies for RESTful APIs, ORM with **Hibernate**, and security enhancements.
- Configured **Apache CXF** within Spring Boot to expose **RESTful** and **SOAP** web services, broadening service interoperability and integration.
- Optimized data fetching using **JPA Entity Graphs** to solve the N+1 selects problem, improving query efficiency and application performance by 20%.
- Configured **Logstash** to parse and enrich logs before loading into **Elasticsearch**, enhancing system monitoring and analysis capabilities by 40%.
- Developed real-time operational dashboards with **Kibana**, improving monitoring and alerting for development teams.
- Implemented **Postman** for comprehensive end-to-end testing, simulating complex user interactions to ensure seamless operation from front-end to back-end.
- Optimized database interactions with an API layer using **Node.js** and **Express**, which improved data retrieval efficiency and reduced server response times.
- Streamlined **JDBC** and **Hibernate** configuration in Spring Boot for robust database management, integrating **SQL Server** and **MongoDB** for relational and NoSQL operations
- Optimized Hibernate settings by setting **SQL Server** dialect, enabling batch processing, and configuring connection pooling with HikariCP for improved database connection efficiency
- Implemented partitioning strategies in **MongoDB** to manage large datasets efficiently, reducing load times
- Enhanced message delivery reliability and system resilience using **Apache Kafka** and **IBM MQ**, configuring appropriate partitions, replication factors, and queues for high availability.
- Optimized **Docker** and **Docker Compose** workflows to streamline development and testing, enhancing container management and deployment processes.

- Managed **Kubernetes** deployments using ConfigMaps, Secrets, and Horizontal Pod Autoscalers, ensuring dynamic configuration and efficient resource scaling.
- Automated CI/CD pipelines using **Jenkins** and **Azure Pipelines**, integrating Git for version control, enabling streamlined builds, tests, and deployments across environments.
- Implemented secure API gateways using **Cloud Functions** and **API Gateway** in GCP, providing scalable and secure endpoints for external integrations.
- Optimized **Google Cloud Functions** for performance and cost-efficiency, leveraging Java runtime and improving function execution times by 25%.
- Orchestrated complex workflows using **Google Cloud Composer**, integrating with Cloud Functions and Kubernetes, which enhanced data processing and workflow automation.
- Automated environment provisioning and teardown using **Terraform** for GCP, streamlining DevOps workflows and ensuring consistent infrastructure setups.
- Integrated Cloud Composer with data orchestration tools like **Apache Airflow**, enhancing workflow scheduling and automation capabilities for large-scale data processing tasks.
- Integrated **Google Cloud Storage** with the Starbucks' data lake, providing secure and scalable storage solutions for unstructured data and improving data accessibility.
- Leveraged **Google Cloud Pub/Sub** for reliable messaging and real-time data streaming between microservices, supporting high-throughput applications with low latency.
- Implemented comprehensive monitoring and logging solutions using Google Cloud Operations Suite (formerly Stackdriver), enabling proactive performance management and reducing downtime by 30%.
- Configured **Azure** infrastructure using **Azure CLI** and **AKS**, setting up **Azure Load Balancers** and **Azure Backup** for high availability and disaster recovery compliance.
- Implemented comprehensive monitoring solutions with **Azure Monitor** and **Application Insights**, coupled with **Log4J** for advanced application logging and performance tracking enhancing performance by 30%.
- Integrated **JIRA** to streamline project management and bug tracking for a multi-team development.
- Managed version control using Git and GitHub, ensuring seamless collaboration among team members.

Environment: Java 11, Node.JS, AEM, Express.JS, Redis, React.JS, Webpack, Babel, JSX, ES6, Tailwind CSS, Styled Components, Redux-Saga, Cypress, Swagger, Spring MVC, Camunda, JPA, Hibernate, Criteria API, Spring, Logstash, Elasticsearch, Kibana, Spring Initializr, JDBC, MySQL, MongoDB, Apache CXF, RESTful, SOAP, Apache Kafka, IBM MQ, Docker, Docker Compose, Kubernetes, Secrets, Horizontal Pod Autoscaler, JIRA, OpenShift, Jenkins, CI/CD, Azure CLI, AKS, Azure Load Balancers, Azure (Backup, Pipelines, Monitor), Google Cloud Pub/Sub, StackDriver, GKE, Google Cloud Functions, Terraform, Application Insights, JUnit, Postman, TestNG, Log4J, Git, JSON, XML, Github.

Role: Java Full Stack Developer

February 2017 to December 2019

Client: TransUnion (Chicago)

Responsibilities:

- Initiated and deployed microservices using **Spring Boot** with **Spring Initializr**, selecting critical dependencies tailored to enhance each service's unique functionality, significantly reducing bootstrap time.
- Deployed **Camunda BPM** for dynamic decision modeling with DMN, enabling real-time decision-making capabilities that aligned with business rules, fostering consistent process outcomes across systems managed by **Camunda**.
- Configured **Camunda** REST APIs for external system interactions, ensuring smooth data flow and operational coherence while enhancing process transparency and reducing integration complexities through **Camunda BPM**'s modular design.
- Optimized database interactions with an API layer using **Node.js** and **Express**, which improved data retrieval efficiency and reduced server response times.
- Established WebSocket communication in **Node.js**, supporting simultaneous two-way user interactions without significant latency issues.
- Developed **Node.js** scripts for backend data processing tasks that interfaced with **MongoDB**, optimizing data handling and processing times by 25%.
- Automated **MongoDB** database migrations and schema updates using custom scripts in a CI/CD pipeline for seamless deployment processes.
- Designed and implemented efficient data storage solutions with **Cassandra**, ensuring robust data availability and consistency across distributed systems.
- Configured a 10-node **Cassandra** cluster, enhancing data availability and load distribution, which reduced response times by 30% during peak access periods.

- Leveraged **Node.js** and **Express.js** to architect and implement scalable microservices, managed RESTful API integrations
- Utilized **Kafka** to architect a high throughput messaging system, facilitating seamless data streaming and real-time analytics across microservices, improving data synchronization and event handling.
- Configured and managed Apache **Kafka** clusters within Spring Boot, setting up detailed topic configurations
- Utilized **Kafka** for message brokering with **Cassandra** for data persistence.
- Utilized **IBM MQ** to architect a high throughput messaging system, facilitating seamless data streaming and real-time analytics across microservices, improving data synchronization and event handling
- Implemented CI/CD pipelines using **Bitbucket** Pipelines to automate code building, testing, and deployment.
- Collaborated with cross-functional teams to review and merge code changes, using **Bitbucket** pull requests.
- Utilized **Bitbucket** branching strategies to maintain code stability and facilitate feature development.
- Selected **Google Cloud Platform** for deploying high-performance big data analytics and machine learning workflows, utilizing **BigQuery** for advanced data warehousing and analytics projects.
- Developed a multi-cloud orchestration layer using **Terraform** to manage resources in both **GCP** and **Azure**, simplifying deployment and scaling of applications, improving deployment efficiency by 25%.
- Customized data synchronization tools using **Python** and APIs from both **GCP** and **Azure** to maintain data consistency across cloud environments.
- Utilized **Azure Application Insights** for advanced performance monitoring and setting custom alerts, resulting in a 15% reduction in mean time.
- Integrated **Azure Security Center** for regular security assessments and compliance, achieving 100% compliance
- Set up dedicated **GitLab Runners** on virtual machines optimized for specific tasks such as high-CPU for compute-intensive jobs and Docker-enabled for jobs requiring containerization
- Enabled Docker within **GitLab Runners** to allow the execution of Docker commands, facilitating container-based builds that closely mirror the production environment.
- Transitioned existing CI pipelines from Docker to **Podman**, modifying scripts in **Jenkins**, **GitLab CI**, and **GitHub Actions**.
- Configured **JIRA** Service Desk to improve IT support interactions, reducing ticket resolution times by 25%
- Streamlined dependency management by setting up **Maven** Central Repository mirroring, which significantly improved build efficiency and reduced internet bandwidth usage.

Environment: Spring Boot, Spring Data, Spring Initializr, Camunda, Microservices, Hibernate, Cassandra, MongoDB, Node.js, Express.js, Apache Kafka, Angular 8, NPM, Grunt, Google Cloud Platform (GCP), BigQuery, Azure, Azure Application Insights, Virtual Private Cloud (VPC), Podman, Jenkins, GitLab CI, GitHub Actions, Azure DevOps, WebSocket, HikariCP, JIRA, Maven.

Role: Java Full Stack Developer

October 2015 to December 2016

Client: Amazon - Hyderabad, India

Responsibilities:

- Developed the User Profile Management Service using **Spring Boot**, seamlessly integrating CRUD operations with AWS RDS (MySQL) to efficiently manage extensive user data
- Configured the Recommendation Service using Spring Boot, employing batch and real-time data processing
- Enhanced system security by integrating Java-based authentication and authorization modules using **Spring Security**, providing secure and robust access control mechanisms for web applications.
- Architected scalable microservices using Spring Boot and Java, focusing on building resilient services
- Advanced the application's loading efficiency by implementing **React's** lazy loading modules, which deferred the loading of non-critical **JavaScript** files until needed, reducing initial load time by up to 25%.
- Implemented the Product Interaction Service in Spring Boot, leveraging **DynamoDB's** scalability to robustly handle millions of user interactions daily.
- Led the migration of legacy applications to Red Hat OpenShift, containerizing applications using Docker and orchestrating deployments with Kubernetes, resulting in a 50% reduction in deployment times.
- Utilized **DynamoDB** for session storage and caching, which facilitated quick data retrieval and scalability under high traffic conditions, improving application responsiveness
- Leveraged **DynamoDB** Streams to capture real-time data modifications, enabling seamless integration with AWS Lambda for automated processing and analytics.

- Implemented fine-grained access control with **DynamoDB**, using **AWS IAM** to secure sensitive data
- Integrated **PostgreSQL**'s advanced replication features to synchronize data across multiple geographical locations, ensuring data consistency and availability.
- Configured PostgreSQL databases to manage extensive product and transaction data, leveraging advanced SQL features to optimize query performance and data integrity
- Integrated **Hibernate** for ORM capabilities with **PostgreSQL**, enhancing data interaction layers by simplifying the management and manipulation of transactional data across over 10,000 daily transactions.
- Leveraged **Node.js** with the **Express** framework to set up **RESTful** API endpoints that efficiently handled requests related to credit scores, user profiles, and financial transactions
- Utilized **Node.js** with **Express** to create scalable and secure RESTful services that efficiently processed thousands of transactions per day, enhancing the reliability of user interactions with the product catalog
- Configured **RabbitMQ** to handle asynchronous message queuing, ensuring reliable message delivery and maintaining data integrity across critical financial transactions
- Employed **RabbitMQ** to process and analyze streaming data in real-time, enabling dynamic decision-making and personalized content delivery on the e-commerce platform
- Leveraged **Redis** for strategic caching to enhance performance by storing frequently accessed data, significantly reducing database load and improving response times by 40%.
- Enhanced application security and user authentication using OAuth 2.0 through AWS API Gateway, ensuring that all data transfers are protected via HTTPS
- Deployed and managed the application using **AWS Elastic Beanstalk**, which facilitated auto-scaling operations and monitored application performance and health via **AWS CloudWatch**.
- Utilized Docker within GitLab CI/CD pipelines for containerization and streamlined deployment processes to AWS ECR and a Kubernetes-managed cluster.
- Monitored detailed application performance and system metrics using JVisualVM and Log4j, for efficient debugging and performance tuning.
- Applied JVM tuning and garbage collection strategies for managing memory efficiently under heavy load
- Applied extensive error handling and validation mechanisms within Spring MVC to secure and stabilize API endpoints

Environment: Java 1.8, Spring Boot, AWS CDK, AWS RDS, PostgreSQL, DynamoDB, Spring Cloud Gateway, Hibernate, JDBC, RabbitMQ, Redis, AJAX, Bootstrap, HTML5, CSS3, JavaScript, TypeScript, React.js, Redux, GitLab CI/CD, Docker, AWS ECR, Kubernetes, JUnit, Mockito, React Testing Library, Jest, Cypress, JMeter, AWS API Gateway, OAuth 2.0, HTTPS, SQL Queries, NoSQL, DAX, Log4j, JVM, JVisualVM, AWS CloudWatch, AWS Elastic Beanstalk

Role: Software Developer

August 2014 to September 2015

Client: Ramco Systems - Hyderabad, Telangana

Responsibilities:

- Conducted thorough analysis of ADP's HRMS architecture, identifying core components like user management and payroll processing, utilizing **Java** and **Oracle** technologies for assessment.
- Optimized Java applications for performance and efficiency, utilizing profiling tools and **JVM** tuning techniques to diagnose and rectify performance bottlenecks
- Enhanced system security by integrating Java-based authentication and authorization modules using Spring Security.
- Optimized front-end performance by minimizing blocking **JavaScript** and **CSS**, using **HTML5** and **CSS3**
- Implemented automated testing frameworks using **Jasmine** for behaviour-driven **JavaScript** testing and **Karma** as a test runner
- Developed meticulous migration roadmap for ADP, prioritizing modules for phased roll-outs during migration to **Microsoft Azure**, showcasing strategic planning and project management skills.
- Provisioned and configured **Azure Virtual Machines**, optimizing compute, memory, and storage
- Implemented a two-phase database transition strategy, migrating 10 terabytes of data from on-premises **Oracle** database to **Azure SQL Database** using **Azure Data Migration Service**.
- Refactored Java-based backend, optimizing session management for cloud compatibility and integrating Azure Managed Identity for secure access.
- Upgraded existing APIs with Azure API Management, enhancing security, caching, and performance.
- Leveraged **Node.js** and **Express.js** for developing microservices, enhancing system responsiveness and scalability.
- Transformed Angular-based front end into a Progressive Web Application (PWA), enhancing offline capabilities and mobile access using **HTML**, **CSS**, and **Angular**.

- Employed **Spring MVC** to streamline request mappings and enhance controller designs, significantly improving response handling and server-side request processing
- Utilized **Spring MVC** to implement Aspect-Oriented Programming (AOP), achieving enhanced modularity and separation of concerns across services, which facilitated better maintenance and scalability
- Configured Hibernate ORM with **Oracle** to optimize database interaction, reducing the complexity of CRUD operations and enhancing database performance.
- Led the HRMS system migration to **Google Cloud Platform (GCP)**, planning and executing the transition with minimal downtime using **Terraform** for infrastructure as code
- Established **Docker** containers for encapsulating the application environment, ensuring consistency across local, development, and production setups, and simplifying cloud deployments.
- Set up Jenkins pipelines for CI/CD, integrated with **SonarQube** for continuous static code analysis
- Configured **Google Cloud Load Balancing** to handle incoming user traffic across multiple instances,
- Implemented comprehensive logging using **Elasticsearch, Logstash, and Kibana (ELK Stack)**
- Utilized GCP's Cloud Armor to manage security policies and protect against DDoS attacks
- Automated performance monitoring with GCP's Stackdriver, providing insights into application health
- Conducted **Kubernetes** orchestration for Docker containers, facilitating scalable deployments

Environment: JavaScript, HTML, Angular, Node.js, Express.js, AJAX, JSON, HTML5, CSS, Oracle Database, CI/CD pipelines, Kubernetes, GCP Cloud Flare, Docker, SonarQube, ELK (Elasticsearch, Logstash, Kibana), GCP Stackdriver, GCP Load Balancer.

EDUCATION

Sathyabama University

August 2010 to May 2014